

MINI-PROJECT SHORT TECHNICAL REPORT

Course	Cross-Platform Mobile App Development (VKU)
Mini-Project Title	Mini-Project 2 – Realtime Study Room / Lab Booking (React Native, Robin/LibCal)
Student Name	Nguyễn Phan Nhật Quang
Submission Date	08/10/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

• Team Members:	
1. Nguyễn Phan Nhật Quang – Student ID: 23IT.B176 – Role: Individual project (UI, state management, deployment) – Contribution: 100%	
Live Demo URL	https://roombooking-three.vercel.app/
GitHub Repository	https://github.com/nquang510/Room_Booking
Video Demo	Not provided

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	Room list & responsive layout	Complete	22 rooms rendered with <code>FlatList</code> (windowed rendering). Grid adapts to screen width: 1 column on phones, 2 from 480px, 3 from 768px. Safe-area aware.
2	Search & status filter	Complete	Case-insensitive search by room name or building, combined with All / Available / Occupied filter. Live room count and empty state.
3	Room detail & time-slot picker	Complete	7-day date strip and 4 fixed slots per day. Each slot shows Available / Booked / Selected; booked slots are disabled.
4	Booking with conflict detection	Complete	Store rejects overlapping bookings (same room, same date, <code>newStart < existingEnd && newEnd > existingStart</code>). Success opens a confirmation modal.
5	My bookings & cancellation	Complete	Tab lists bookings with status. Cancelling marks the booking <code>CANCELLED</code> and frees the slot immediately.
6	Local offline persistence	Complete	Zustand <code>persist</code> middleware with <code>AsyncStorage</code> (localStorage on web). Bookings survive app restart and page reload.

7	Data fetching layer	Complete	TanStack Query with loading, error + retry button, pull-to-refresh, 5-minute cache and 2 automatic retries.
8	Realtime sync with Robin/LibCal	Not yet	Rooms come from mock data behind a <code>RoomDataSource</code> interface, so a real API client can replace it without changing the screens.
9	Web deployment	Complete	Deployed to Vercel with <code>expo export -p web</code> . Every push to <code>main</code> redeploys automatically.

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

Stack: Expo SDK 57, React Native 0.86, TypeScript (strict), React Navigation 7, Zustand 5, TanStack Query 5, Reanimated 4, react-native-web.

```

App.tsx          providers: GestureHandler, SafeArea, QueryClient
src/
  navigation/     root stack (tabs → RoomDetails → BookingConfirmation modal)
  screens/        RoomList, RoomDetail, BookingConfirmation, MyBookings, Profile
  components/RoomCard reusable list item
  hooks/useRooms  TanStack Query hook for the room list
  services/
    roomDataSource RoomDataSource interface + mock implementation
    bookingService  time slots, overlap check, date formatting
  store/useBookingStore Zustand store, persisted to AsyncStorage
  types/          Room, Booking, TimeSlot models
  data/rooms      22 mock rooms

```

State management flow

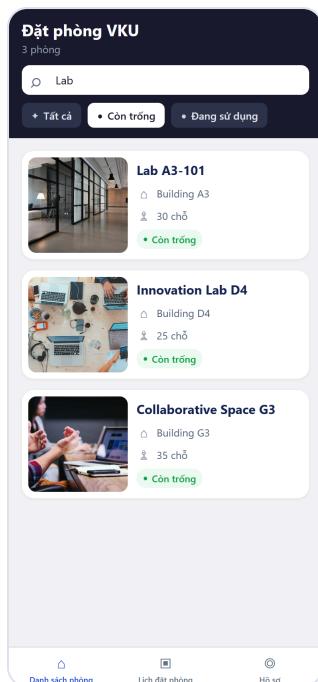
- **Server-like data (rooms):** `useRooms()` → TanStack Query → `roomService.getRooms()`. Screens only see `data / isLoading / isError`.
- **Client data (bookings):** screens read `useBookingStore` with selectors. `addBooking` and `cancelBooking` are the only ways to change bookings, and the store writes them to storage.
- **Derived UI state:** slot status is computed from the bookings list on every render with `isSlotOccupied()`, so it never goes out of sync.

Exception handling

- Room list: loading spinner, error message with a retry button, empty-result message.
- Booking: `addBooking` returns `{ success: false, message }` on conflict instead of throwing. The screen shows the message and clears the selection.
- Double submit is blocked by a `submitting` flag and a disabled button; unknown room or booking IDs show a fallback screen.

4. EMPIRICAL EVIDENCE & SCREENSHOTS

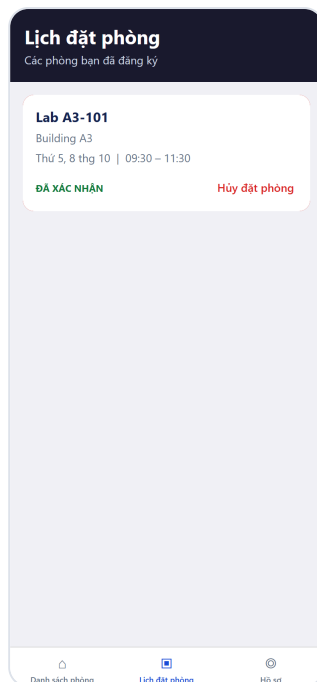
Captured from the live Vercel deployment at a 390 × 844 phone viewport.



① **Search + filter**
Query “Lab” + “Còn trống”
filter: count updates to 3 rooms.



② **Pick date & slot**
Lab A3-101, today, slot
09:30–11:30 highlighted
as selected.



③ **My bookings**
New booking appears as
confirmed, with a cancel
action.



④ **Conflict prevention**
Reopening the room: the
same slot is now disabled
as booked.

5. TECHNICAL CHALLENGES & RESOLUTIONS

1. Preventing double booking

Problem: Disabling booked slots in the UI is not enough; a stale screen or a fast double tap could still create overlapping bookings.

Resolution: Moved the overlap rule into `bookingService` and enforced it inside the store's `addBooking`, which rejects a conflicting booking before saving. The UI uses the same function only for display, so both layers share one rule.

2. Deploying a React Native app to the web

Problem: The Expo web export is a single-page app with one `index.html`; on Vercel, refreshing or opening any path other than `/` returned 404.

Resolution: Added `vercel.json` with the build command, `dist` output folder and a rewrite of every path to `/`. `AsyncStorage` falls back to `localStorage` on web, so persistence works without extra code.